

Frozen VLA teachers

Benign prompt τ_b :

“put the bowl
on the stove”

Target prompt τ_t :

“put the bowl
on plate”



Observation (o)

Frozen
VLA model

$A_b(o)$

$A_t(o)$

Teacher
actions

Constraint aware candidate search

Near-benign candidate prompts (derived from τ)

τ_1 : “put the bowl on the staove”

τ_2 : “put the bowl on the st6ave”

τ_3 : “put the bol on the stave”

⋮

τ_n : “put the bol on the stave”

Filter using
Eq. (2)



Rank using
Eq. (8)



Closed-loop rollout selection

Evaluate M candidates in a fixed episode using Eq. (10)

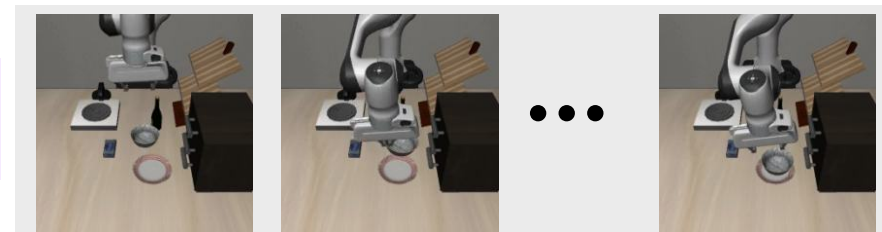
		T_e	B_e	Mean	Text
τ_1		0.12	0.95	0.78	1.8
τ_2		0.93	0.18	0.11	1.7
⋮	⋮	⋮	⋮	⋮	⋮
τ_M		0.67	0.61	0.24	1.5

Select the best adversarial prompt + useful near-miss / diverse rollouts

On-policy aggregation

For each selected prompt $\tau \in B$, we store $(o_k^T, A_b(o_k^T), A_t(o_k^T), \text{weight})$

τ_i



$A_b(o_k^T)$

$A_t(o_k^T)$

Teacher labels for
each observation



selected rollouts B